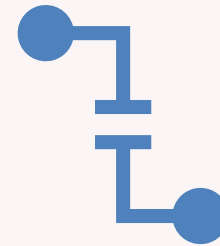


LESSON 6



BOOK CHAPTER 25



CAPACITANCE

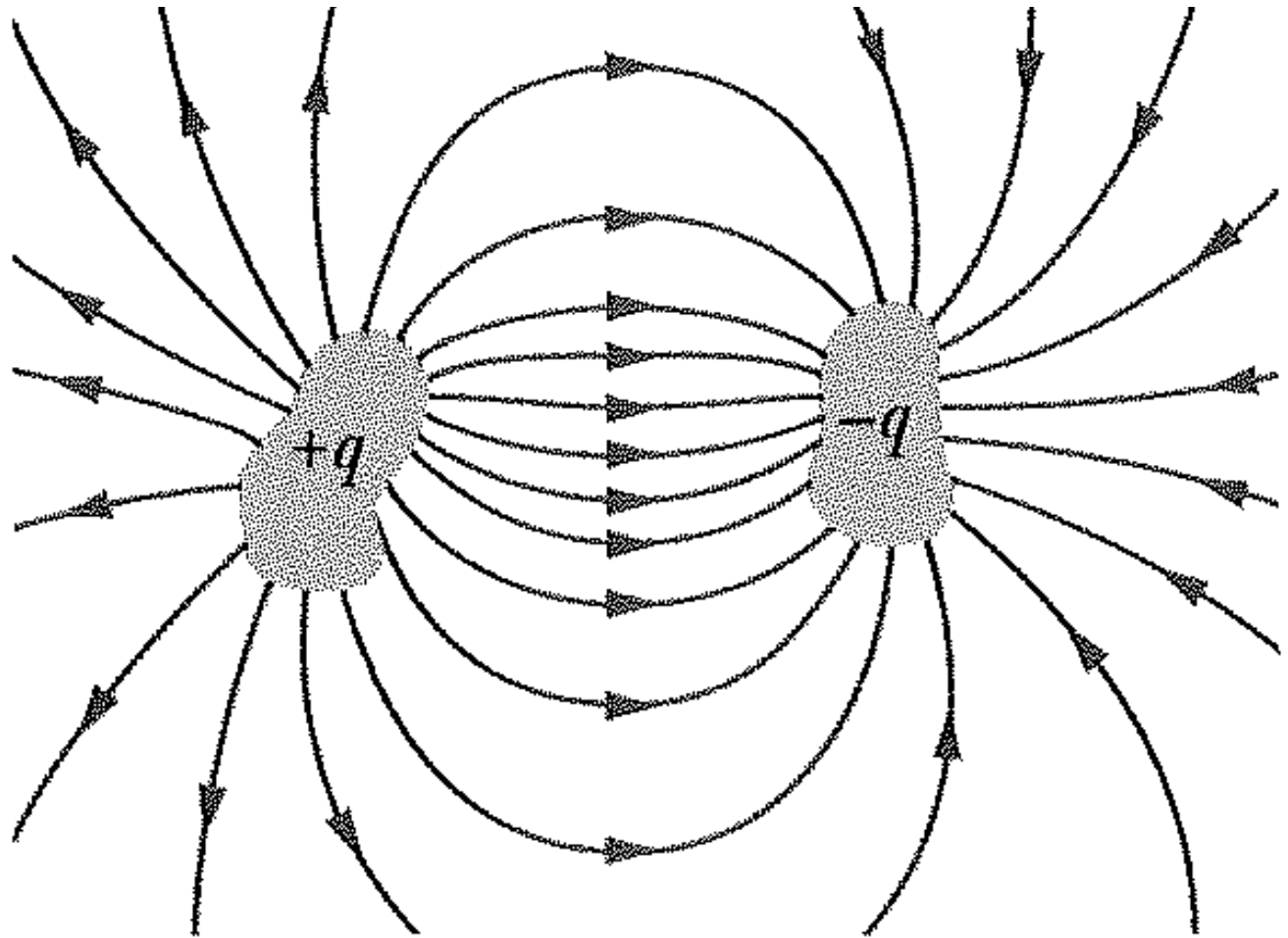
CAPACITANCE



An assortment of capacitors.

Paul Silvermann/Fundamental Photographs

Capacitor:



Two conductors, isolated electrically from each other and from their surroundings, form a **capacitor**. When the capacitor is charged, the charges on the conductors, or *plates* as they are called, have the same magnitude q but opposite signs.

Capacitance

A capacitor consists of two isolated conductors (the plates) with charges $+q$ and $-q$.

The charge q and the potential difference V for a capacitor are proportional to each other; that is,

$$q \propto V$$

Therefore,

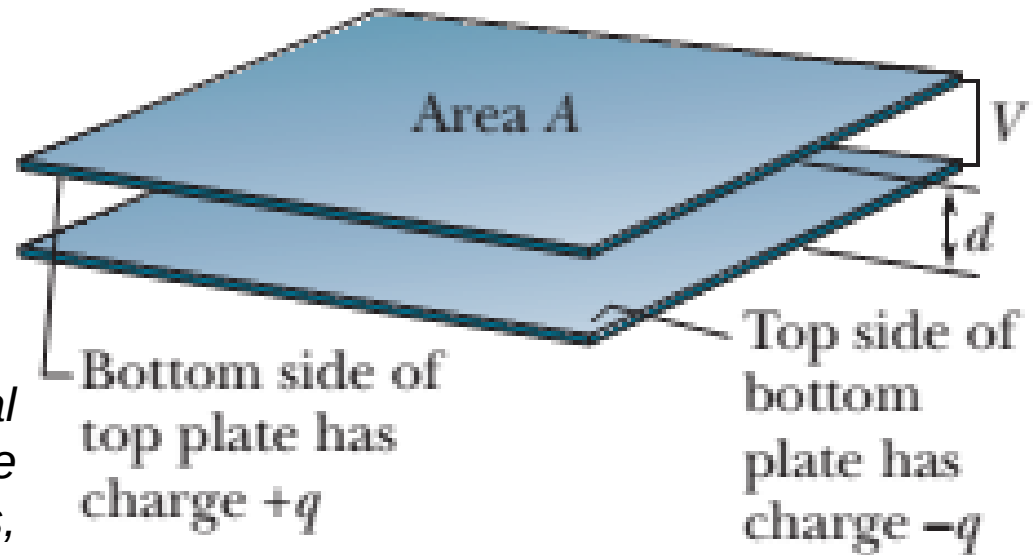
$$q = CV$$

The proportionality constant C is called the capacitance of the capacitor.

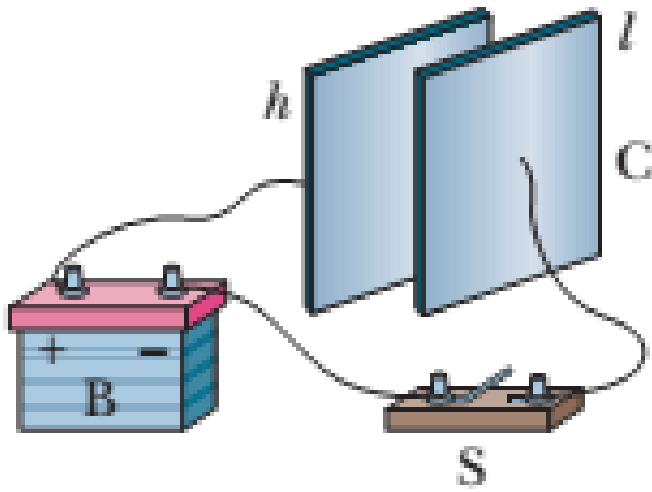
The value of C depends only on the geometry of the plates and *not* on their charge or potential difference. The capacitance is a measure of how much charge must be put on the plates to produce a certain potential difference between them: The *greater the capacitance, the more charge is required*.

$$C = \frac{q}{V}$$

The SI unit of capacitance is the coulomb per volt. Common name is Farad (F):



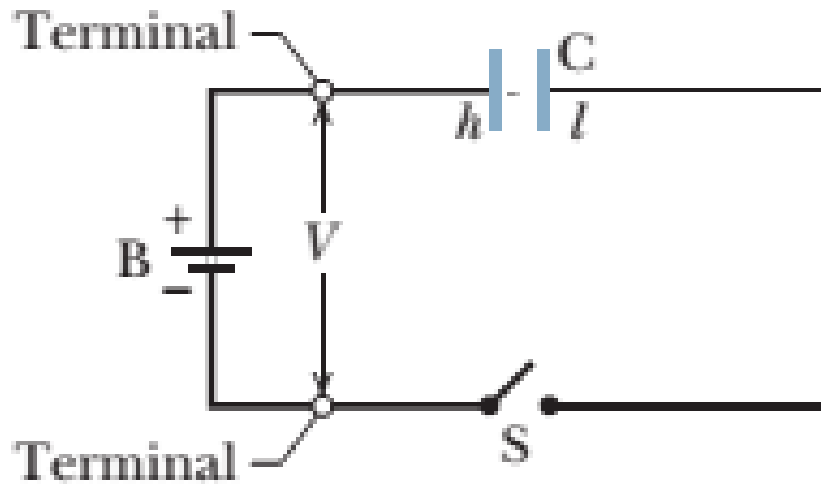
Charging a Capacitor:



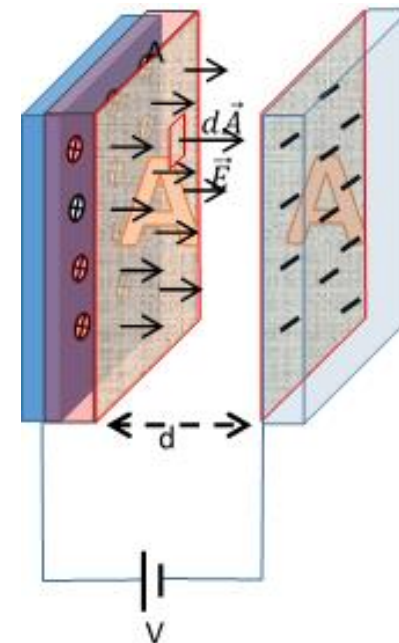
(a)

(a) Battery B, switch S, and plates h and l of capacitor C, connected in a circuit.

(b) A schematic diagram with the *circuit elements* represented by their symbols.



(b)



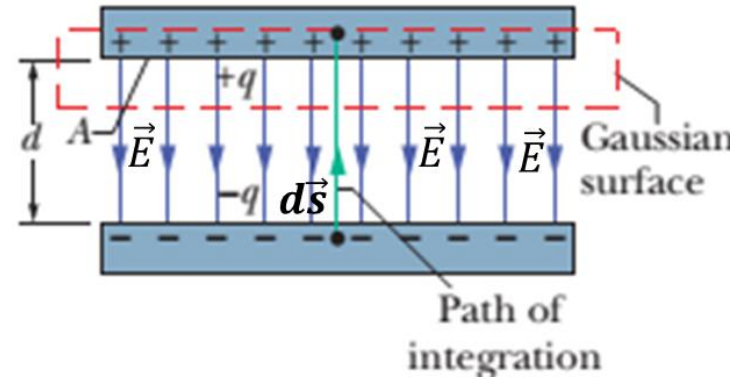
Parallel plate capacitor

Calculating the Capacitance: A parallel-Plate Capacitor:

Applying Gauss' Law:

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q$$

Here q is the charge enclosed by a Gaussian surface and $\oint \vec{E} \cdot d\vec{A}$ is the net electric flux through that surface.



$$\epsilon_0 \oint E(dA) \cos 0^\circ = q \quad \left| \quad \begin{array}{l} \text{Since} \\ \vec{E} \text{ and } d\vec{A} \text{ are parallel} \end{array} \right.$$

Therefore, $\epsilon_0 EA = q$

A is the area of that part of the Gaussian surface through which there is a flux.

The potential difference between the plates of a capacitor is related to the field $\vec{E}$ by

$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s} = - \int_-^+ E \cos 180^\circ ds$$

$$V = \int_-^+ E ds \quad \text{Where, } V = V_f - V_i$$

$$V = E \int_{s=0}^{s=d} ds = Ed$$

We have $C = \frac{q}{V} = \frac{\epsilon_0 EA}{Ed} = \frac{\epsilon_0 A}{d}$

$$C = \frac{\epsilon_0 A}{d}$$

A Spherical Capacitor:

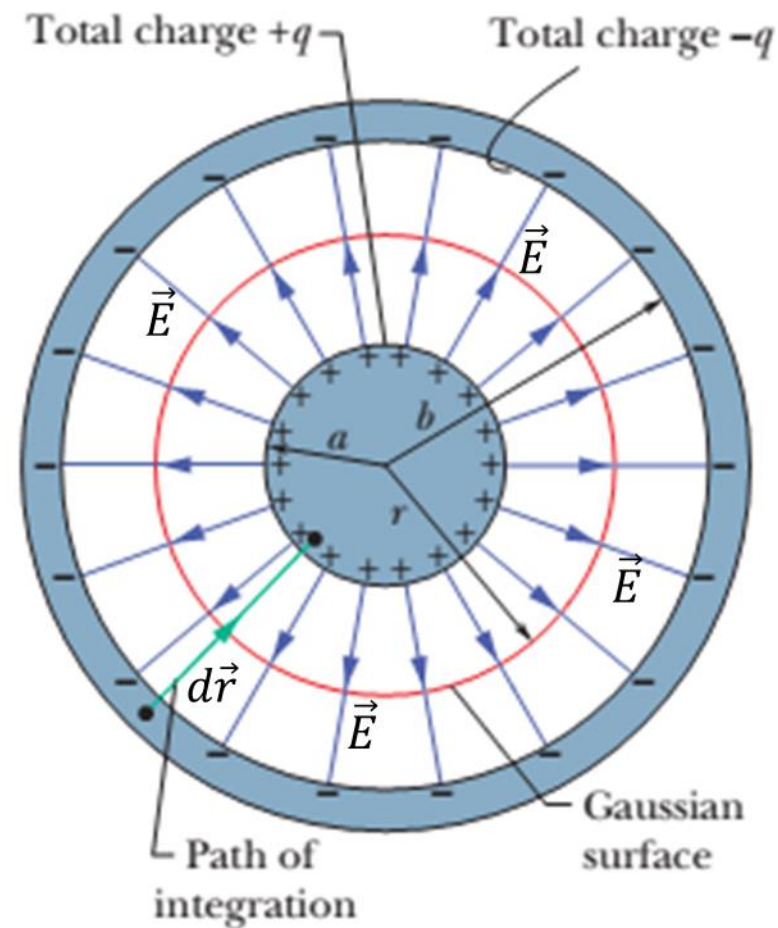


Figure : A central cross section of a capacitor that consists of two concentric spherical shells, of radii a and b .

The adjacent figure shows a central cross section of a spherical capacitor that consists of two concentric spherical shells, of radii a and b .

As a Gaussian surface we draw a sphere of radius r concentric with the two shells. Then

Applying Gauss' law $\epsilon_0 EA = q$

$\epsilon_0 E(4\pi r^2) = q$ Where, $A = 4\pi r^2$ is the area of the spherical Gaussian surface.

$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

We know

$$V = \int_{-}^{+} E ds = - \int_{-}^{+} E dr \quad [\text{since } ds = -dr]$$

$$V = -\frac{q}{4\pi\epsilon_0} \int_{r=b}^{r=a} \frac{dr}{r^2} = \frac{q}{4\pi\epsilon_0} \left(\left| \frac{1}{r} \right|_{r=b}^{r=a} \right)$$

$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right) = \frac{q(b-a)}{4\pi\epsilon_0 ab}$$

$$C = \frac{q}{V} = \frac{4\pi\epsilon_0 ab}{b-a}$$

An isolated sphere:

We can assign a capacitance to a *single* isolated spherical conductor of radius R by assuming that the "missing plate" is a conducting sphere of infinite radius.

To find the capacitance of the conductor, we first rewrite $C = \frac{4\pi\epsilon_0 ab}{b-a}$ as

$$C = \frac{4\pi\epsilon_0 a}{1 - \frac{a}{b}} \quad [\text{Dividing both numerator and denominator by } b]$$

If we then let $b \rightarrow \infty$ (infinity), we get

$$C = \frac{4\pi\epsilon_0 a}{1 - 0} = 4\pi\epsilon_0 a$$

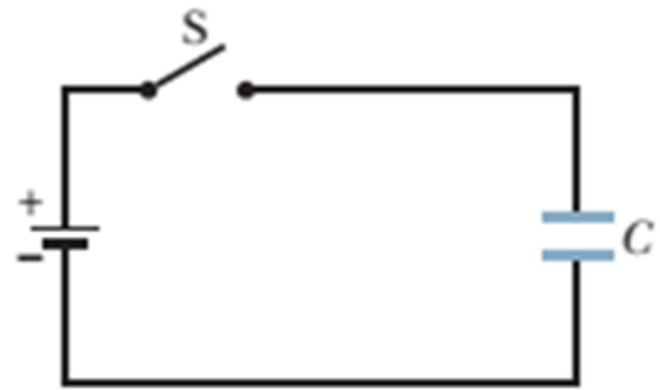
By substituting R for a , we get

$$C = 4\pi\epsilon_0 R$$

Which is the capacitance for isolated sphere.

Problem 2 (Book chapter 25):

The capacitor in the adjacent Fig. has a capacitance of $25 \mu F$ and is initially uncharged. The battery provides a potential difference of 120 V. After switch S is closed, how much charge will pass through it?



Answer:

We know

$$q = CV$$

$$q = 25 \times 10^{-6} \times 120 = 3 \times 10^{-3} \text{ C}$$

Given

$$C = 25 \mu F = 25 \times 10^{-6} F$$

$$V = 120 \text{ V} \quad q = ?$$

Problem 3 (Book chapter 25):

A parallel-plate capacitor has circular plates of 8.20 cm radius and 1.30 mm separation. (a) Calculate the capacitance. (b) Find the charge for a potential difference of 120 V.

Answer:

(a) We know

$$C = \frac{\epsilon_0 A}{d} = \frac{\epsilon_0 (\pi r^2)}{d}$$

$$C = \frac{8.854 \times 10^{-12} \times 3.1416 \times (0.082)^2}{1.3 \times 10^{-3}} = 143.87 \times 10^{-12} \text{ F}$$

Given

$$r = 8.20 \text{ cm} = 0.082 \text{ m}$$

$$d = 1.30 \text{ mm} = 1.3 \times 10^{-3} \text{ m}$$

$$V = 120 \text{ V}$$

(b) We know

$$q = CV = 143.87 \times 10^{-12} \times 120 = 17.26 \times 10^{-9} \text{ C}$$

Problem 4 (Book chapter 25):

The plates of a spherical capacitor have radii 38.0 mm and 40.0 mm. (a) Calculate the capacitance. (b) What must be the plate area of a parallel-plate capacitor with the same plate separation and capacitance?

Answer:

(a) We know

$$C = \frac{4\pi\epsilon_0 ab}{b - a} = \frac{1}{9 \times 10^9} \frac{38 \times 40 \times 10^{-6}}{2 \times 10^{-3}}$$

$$C = 84.44 \times 10^{-12} \text{ F}$$

Given

$$a = 38 \text{ mm} = 38 \times 10^{-3} \text{ m}$$

$$b = 40 \text{ mm} = 40 \times 10^{-3} \text{ m}$$

$$b - a = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$$

(b) For a parallel plate capacitor, we know

$$C = \frac{\epsilon_0 A}{d}$$

$$A = \frac{Cd}{\epsilon_0} = \frac{84.44 \times 10^{-12} \times 2 \times 10^{-3}}{8.854 \times 10^{-12}}$$

$$A = 19.074 \times 10^{-3} \text{ m}^2$$

Here

$$d = b - a = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$$

$$A = ?$$

Problem 6 (Book chapter 25):

You have two flat metal plates, each of area 1.00 m^2 , with which to construct a parallel-plate capacitor. (a) If the capacitance of the device is to be 1.00 F , what must be the separation between the plates? (b) Could this capacitor actually be constructed?

Answer:

(a) We know

$$C = \frac{\epsilon_0 A}{d}$$

$$d = \frac{\epsilon_0 A}{C} = \frac{8.854 \times 10^{-12} \times 1}{1} = 8.854 \times 10^{-12} \text{ m}$$

Given

$$A = 1.00 \text{ m}^2$$

$$C = 1.00 \text{ F}$$

(a) $d = ?$

(b) Could this capacitor actually be constructed?

(b)

It is not possible to construct a capacitor by the separation distance, $d = 8.854 \times 10^{-12} \text{ m}$, because d value is less than the minimum size of an atom.

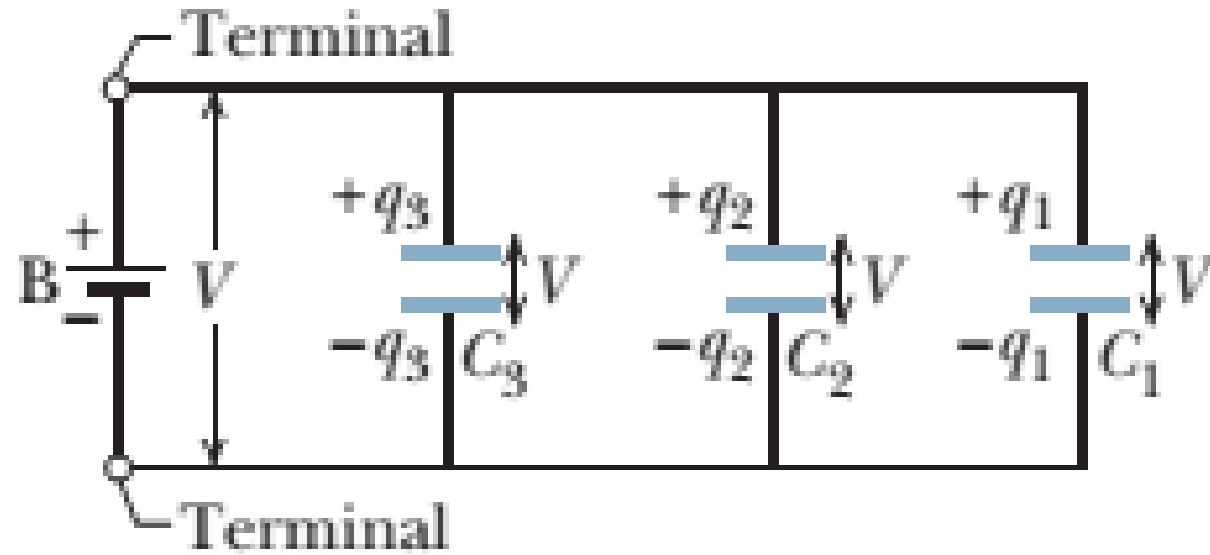
Capacitors in parallel combination:

Charge on each capacitor:

$$q_1 = C_1 V$$

$$q_2 = C_2 V$$

$$q_3 = C_3 V$$



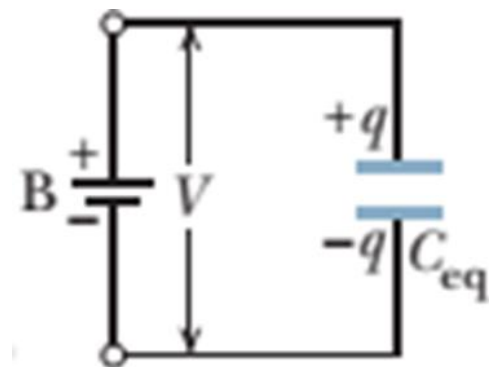
The total charge on the parallel combination is then

$$q = q_1 + q_2 + q_3 = (C_1 + C_2 + C_3)V$$

The equivalent capacitance, with the same total charge q and applied potential difference V as the combination, is then

$$C_{eq} = \frac{q}{V} = \frac{(C_1 + C_2 + C_3)V}{V} = C_1 + C_2 + C_3$$

$$C_{eq} = C_1 + C_2 + C_3$$





Thank You